

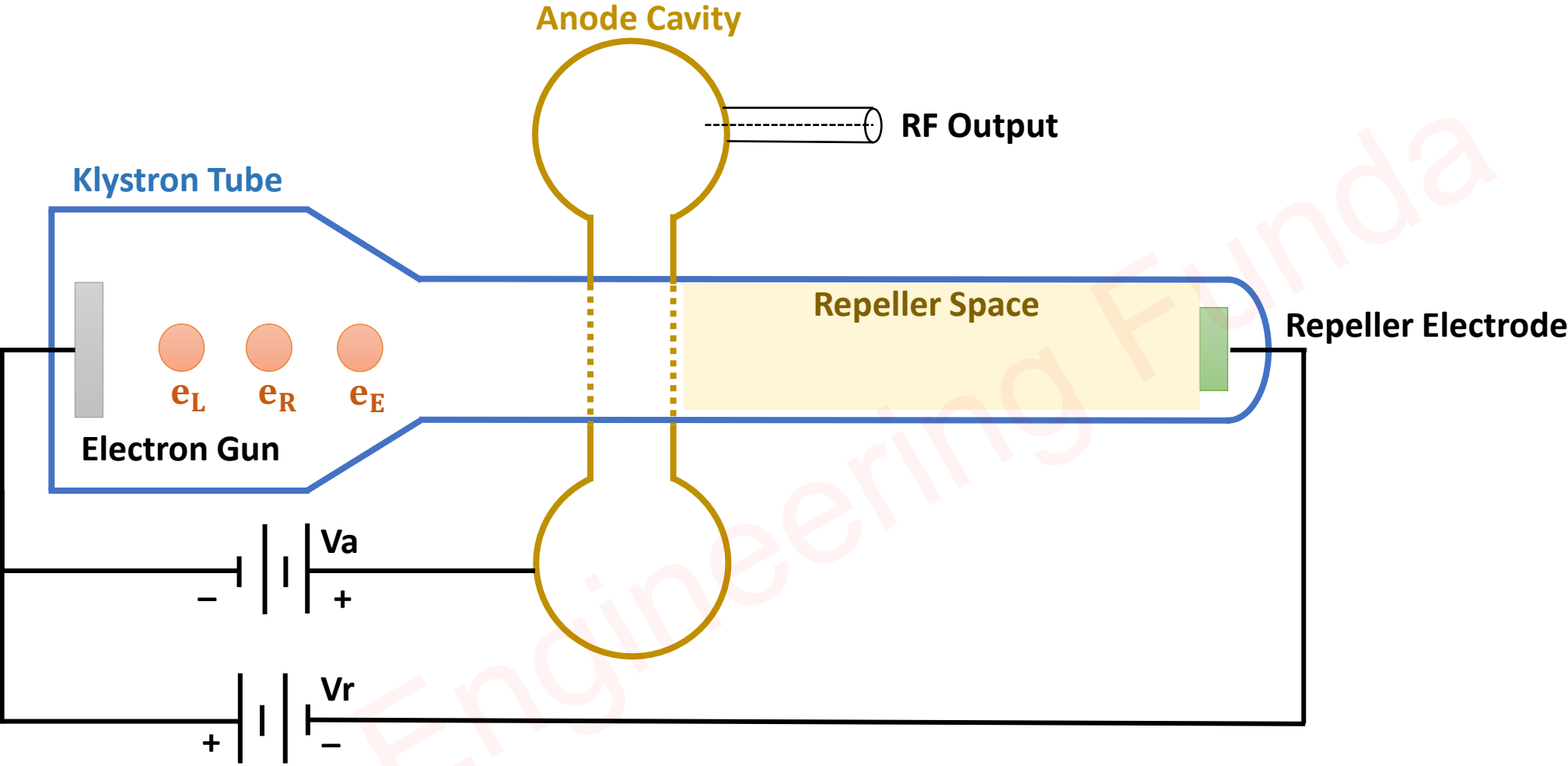
Reflex Klystron

Basics of Reflex Klystron

- ❑ Reflex Klystron is a Microwave Oscillator.
- ❑ A Reflex Klystron is a Vacuum Tube that generates microwave frequencies.
- ❑ Reflex Klystron operates on the principle of velocity modulation of an electron beam.
- ❑ Mainly it is used in the Laboratory as a Microwave Source and It is also used in RADARs.

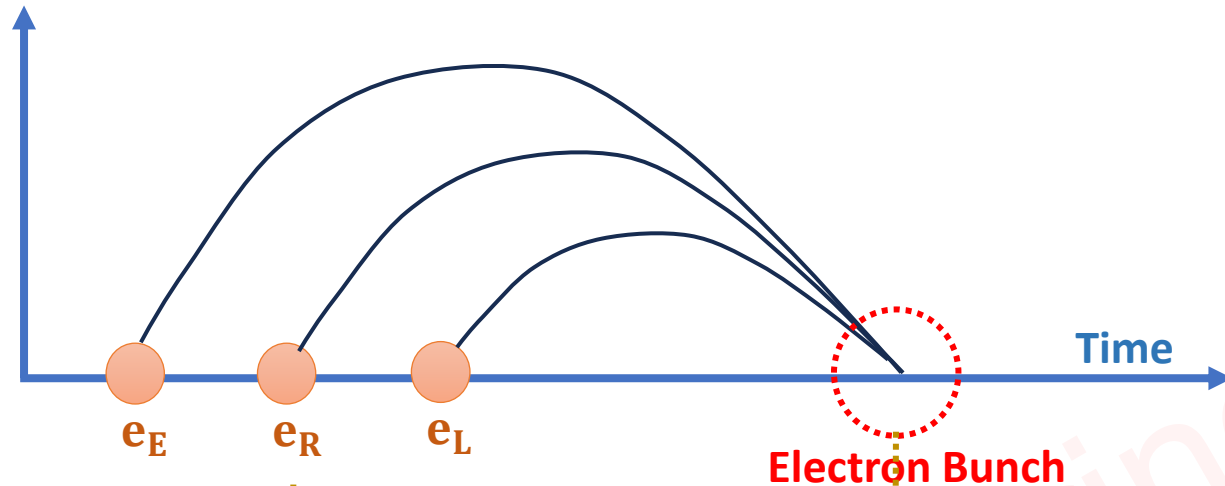


Structure of Reflex Klystron



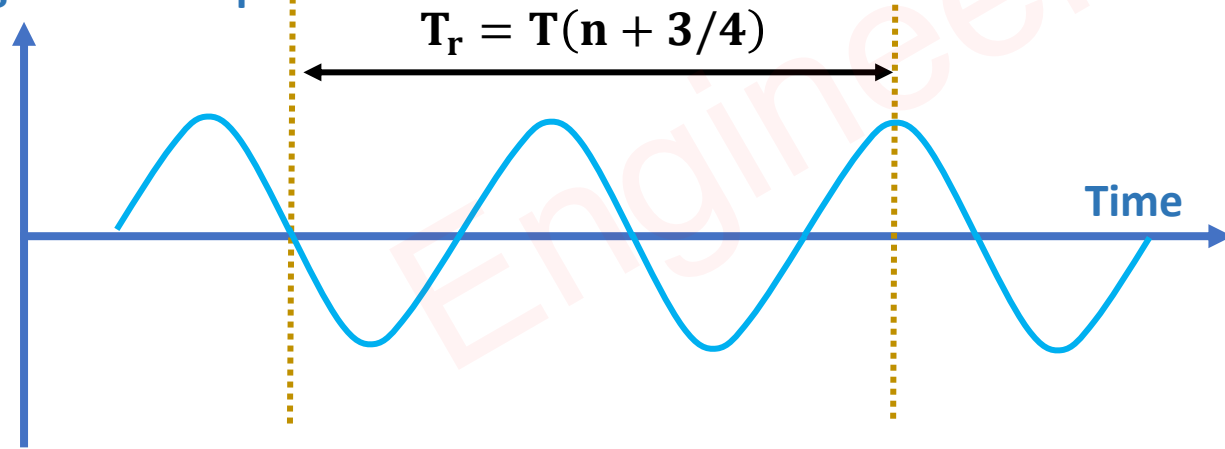
Applegate Diagram of Reflex Klystron

Repeller Space



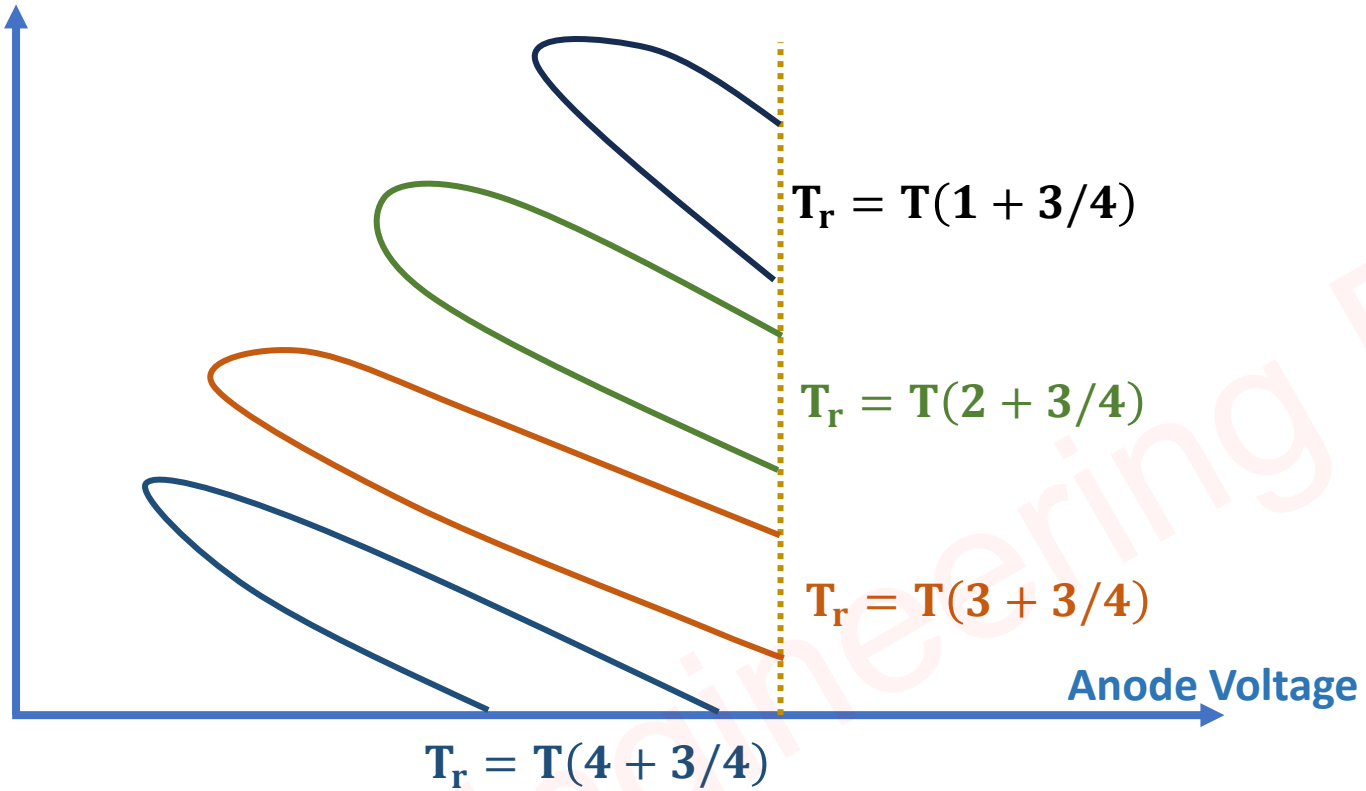
- ❑ Early Electrons travels higher distance in Repeller Space due to higher velocity which takes more time to arrive at Anode Cavity.
- ❑ Late Electrons travels Lower distance in Repeller Space due to Lower velocity which takes Less time to arrive at Anode Cavity.
- ❑ This velocity modulation is done by Repeller Voltage.

Voltage Across Gap



Characteristics of Reflex Klystron

Repeller Voltage



Performance Parameters of Reflex Klystron

- Frequency Range: 1 to 50GHz
- Output Power: 1mW to few MW
- Theoretical Efficiency is 22.78% but practically it will be there in the range of 10% to 20%.

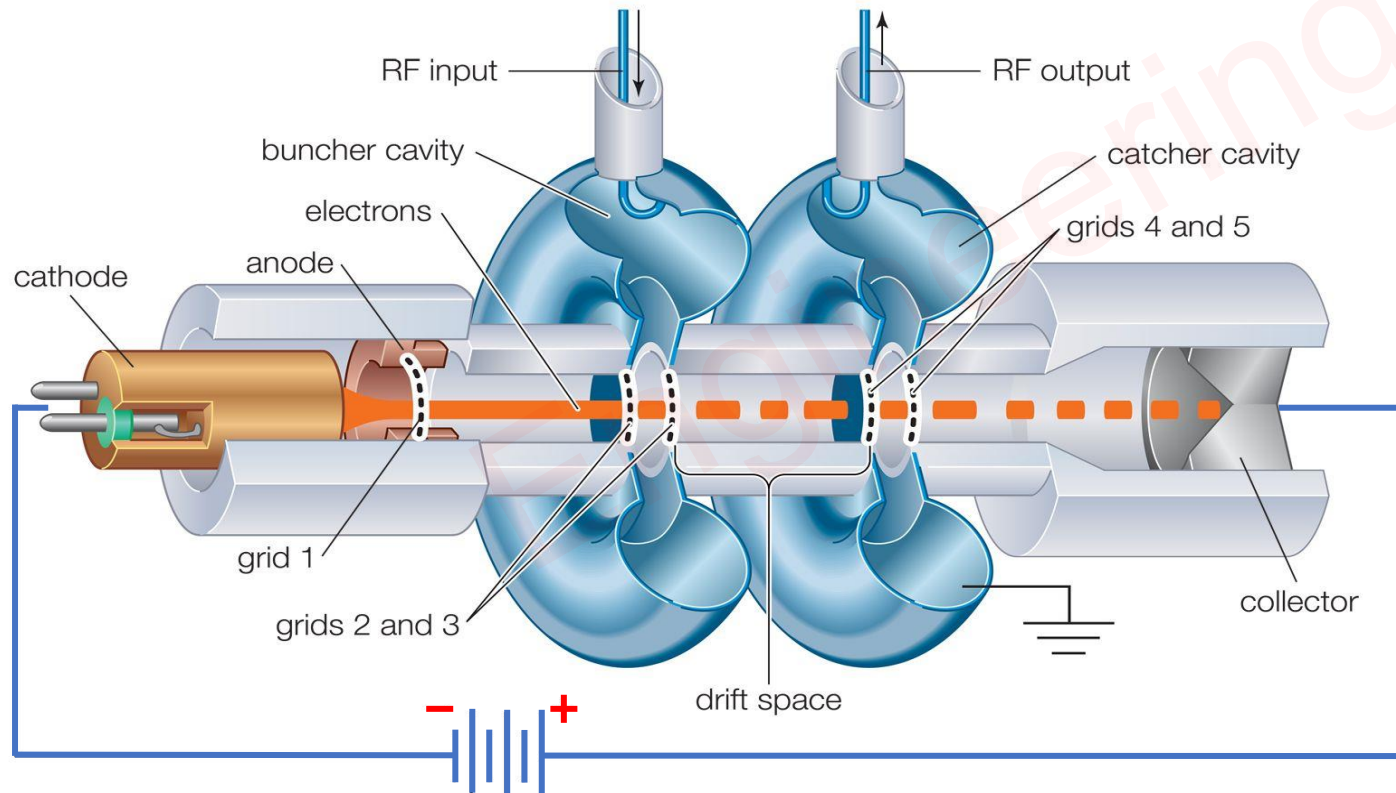
Applications of Reflex Klystron

- Microwave Source
- RADARs
- Communication Systems
- Laboratory Test Bench

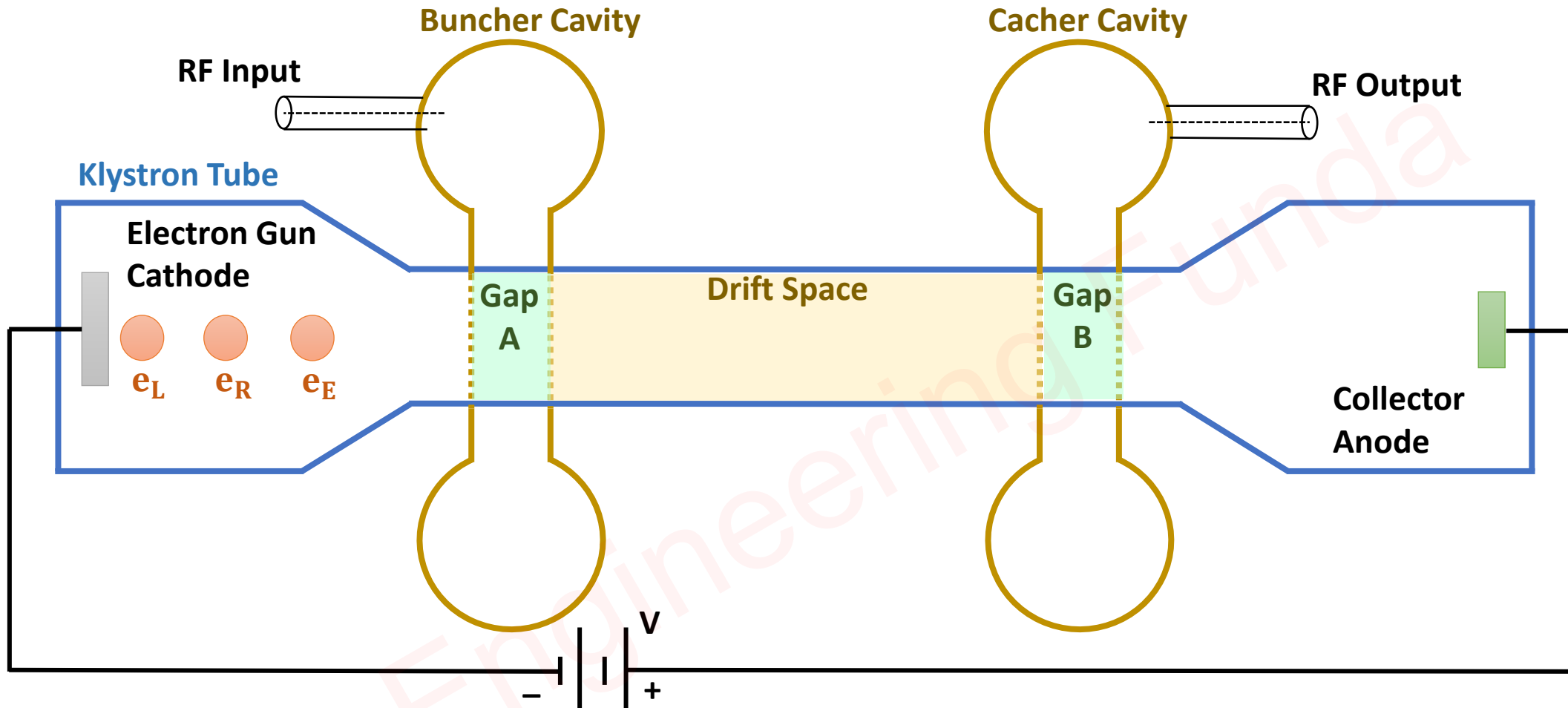
Two Cavity Klystron Amplifier

Basics of Two Cavity Klystron Amplifier

- ❑ Two Cavity Klystron is an Amplifier that amplifies RF Signal.
- ❑ Two Cavity Klystron Amplifier Functions based on the principle of Velocity Modulation.
- ❑ It consists of Two Cavities: Buncher Cavity and Cacher Cavity.
- ❑ RF Input is Buncher Cavity and Amplified RF Output is carried out from Cacher Cavity.



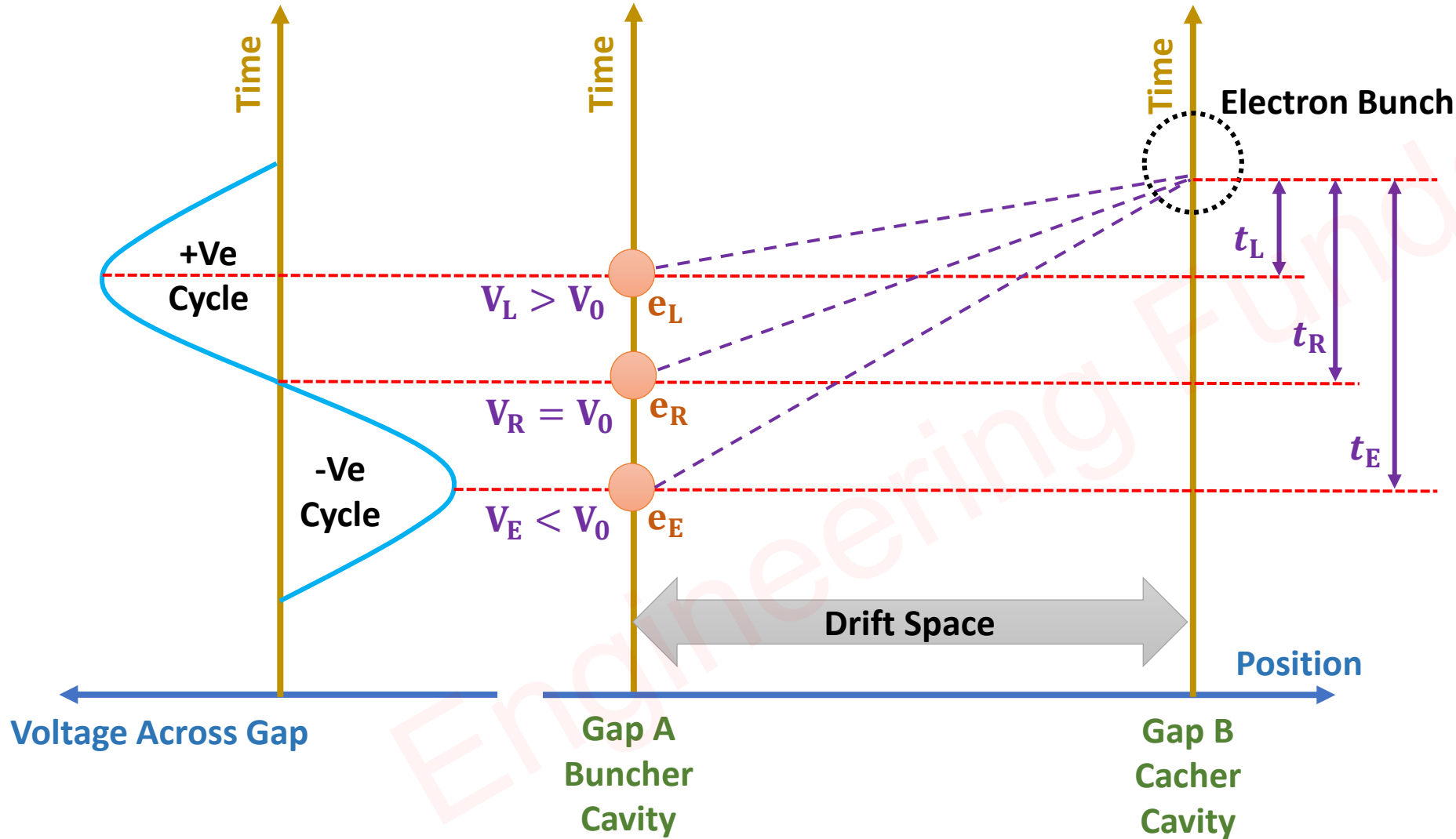
Structure of Two Cavity Klystron Amplifier



□ Without RF Signal, All the electrons travels with same velocity towards Anode.

$$\Rightarrow \frac{1}{2}mv_0^2 = eV \Rightarrow v_0 = \sqrt{\frac{2eV}{m}}$$

Applegate Diagram of Two Cavity Klystron Amplifier



Applications of Two Cavity Klystron Amplifier

- It offers high Gain, so it is used in high-power RF Applications.
- It is used in High Power Pulsed RADAR System
- It is also used in Satellite Communications

Engineering Funda

Reflex Klystron Vs Two Cavity Klystron

Parameters	Reflex Klystron	Two Cavity Klystron
Number of Cavity	❖ One Cavity	❖ Two Cavities: Buncher Cavity and Cacher Cavity
Device Type	❖ Oscillator	❖ Amplifier
Power	❖ Generates Relatively low Power	❖ Operates at High Power
Velocity Modulation	❖ It done by Repeller Electrode	❖ It is done by RF Input at Buncher Cavity
Frequency Range	❖ 1GHz to 50GHz, It has higher frequency stability.	❖ 2GHz to 12GHz, At higher frequencies, it has issues with frequency stability.
Operating Mode	❖ Operates with Continues wave and Pulsed wave mode	❖ Mostly Operates with Continues wave
Structure Complexity	❖ Simple and Small	❖ Complex and Large

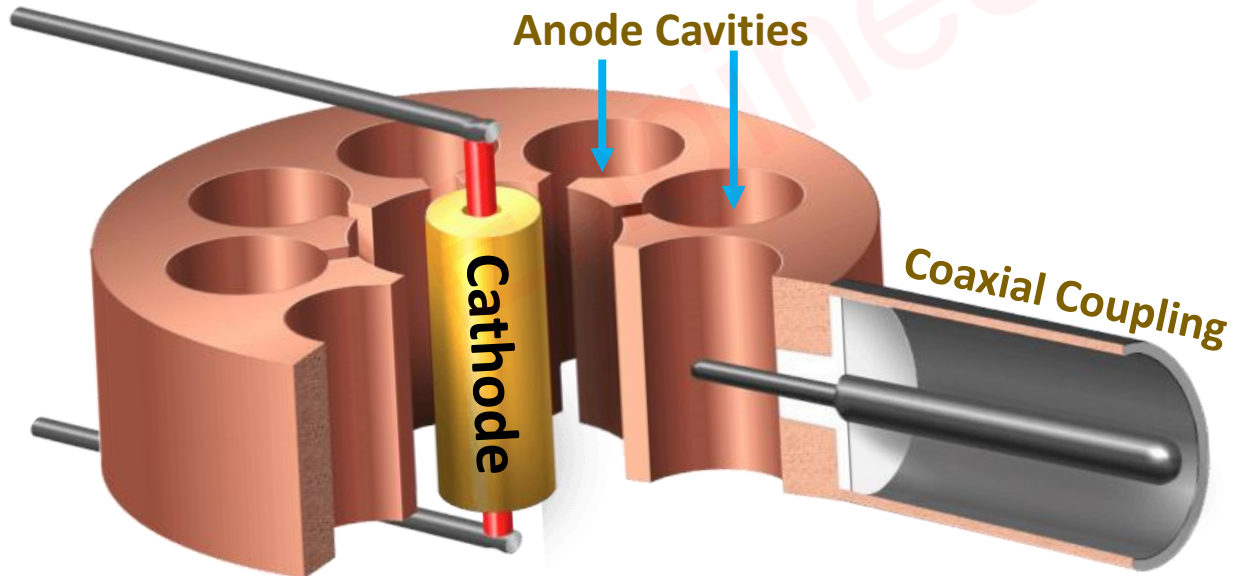
❖ Summary

- ❑ Reflex Klystron is simpler, compact, and easier to tune but limited in power and efficiency.
- ❑ Two Cavity Klystron is more complex, capable of higher power and efficiency, and suitable for amplification, but offers less frequency flexibility.
 - These characteristics make Reflex Klystrons ideal for low-power, tunable applications and Two Cavity Klystrons suitable for high-power amplification needs.

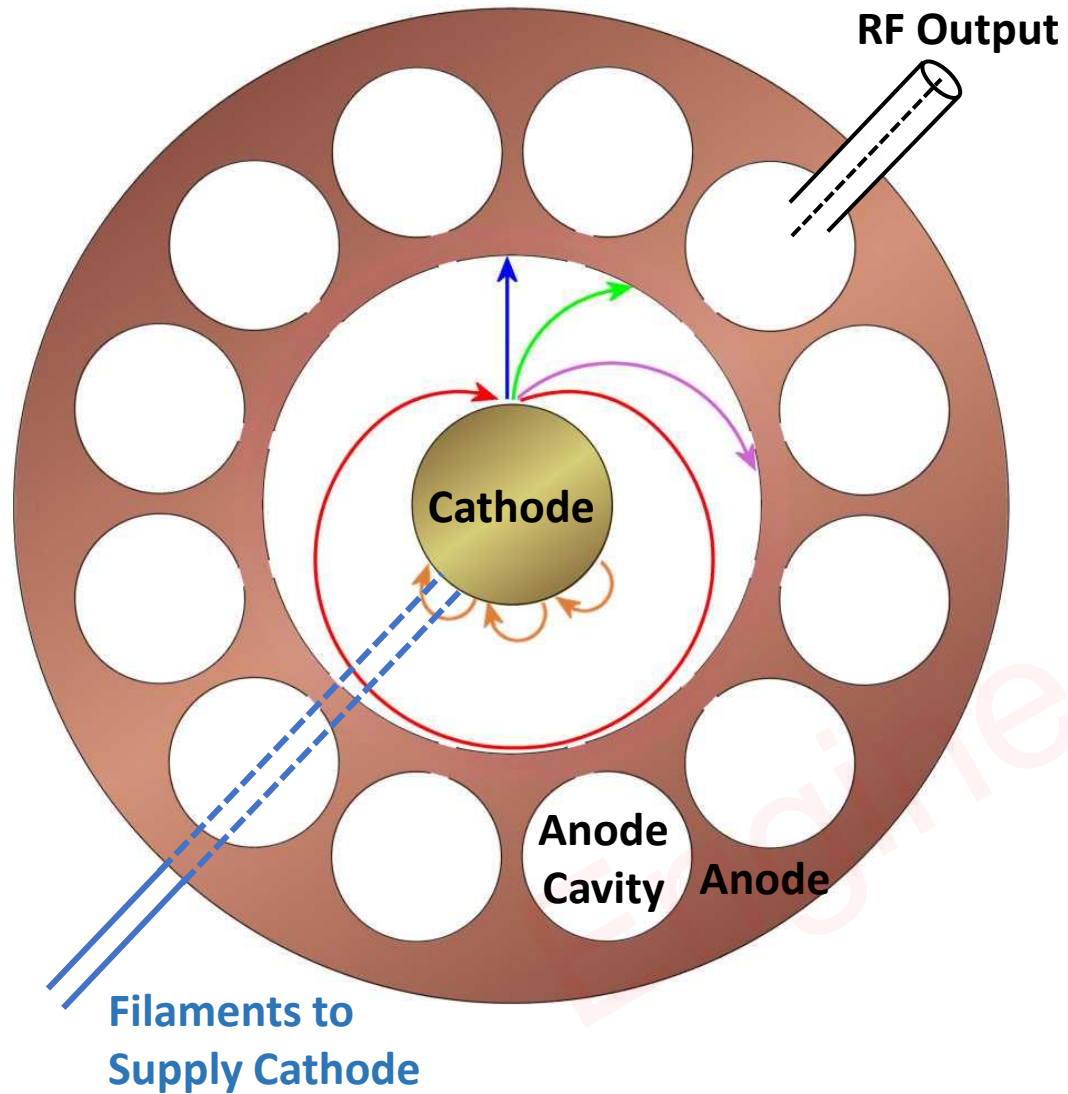
Magnetron Oscillator

Basics of Magnetron Oscillator

- ❑ Magnetron Oscillator is a type of microwave tube that generates high-frequency microwave oscillations.
- ❑ Magnetron Oscillator is Multi Cavity Device available with 8 to 20 Cavities.
- ❑ Magnetron Oscillator works with fixed frequency based on structure and is available with a frequency range from 0.6GHz to 30GHz.
- ❑ Magnetron Oscillator is widely known for its use in radar systems, microwave ovens, and various communication devices.

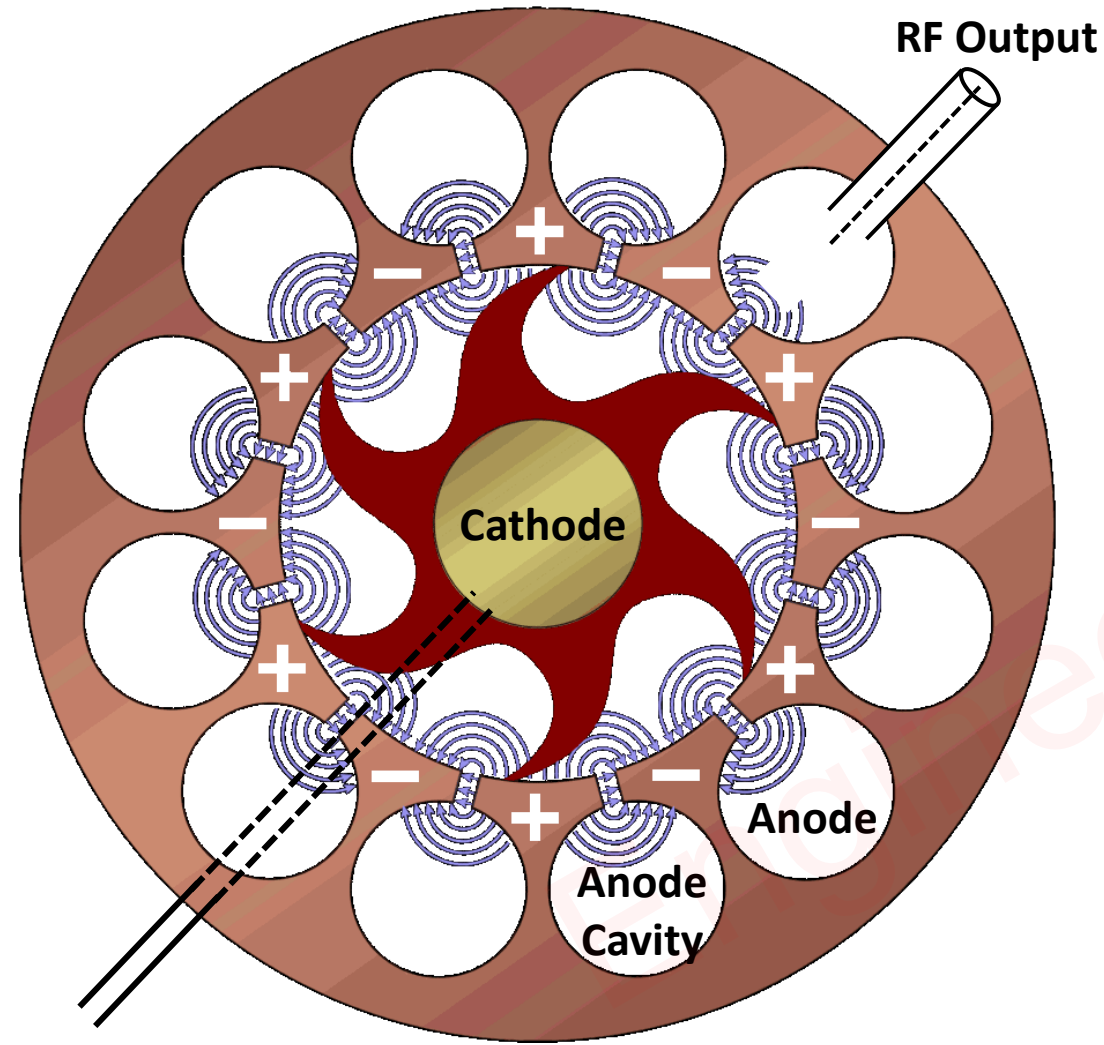


Structure of Magnetron Oscillator



- ❑ Magnetron Oscillator is Multi Cavity Device available with 8 to 20 Cavities. Adjacent Cavity is having a 180-degree shift with each other, These cavities are Anode.
- ❑ Filaments are connected with Cathode to generate electrons.
- ❑ Strong Magnet is placed generating Magnetic field {Direction of H field is coming out of the screen} .
- ❑ Output Coupling is done with the use of a Coaxial Loop. It can be done with the use of Waveguide.
- ❑ Without a Magnetic Field Electrons will go from Cathode to Anode directly.
- ❑ Due to the Magnetic Field, Electrons will follow a curved path.

Working of Magnetron Oscillator



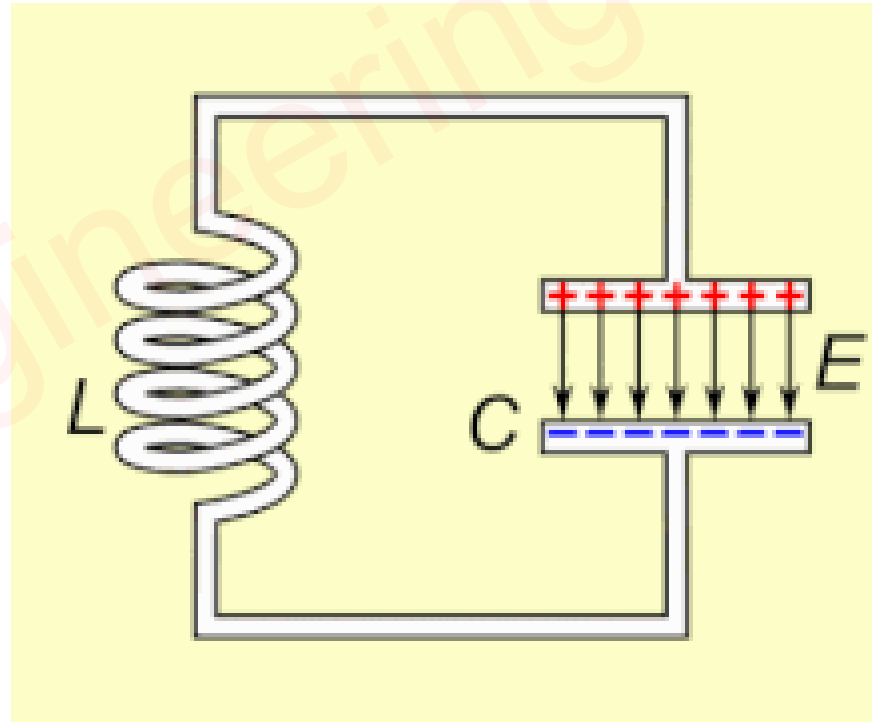
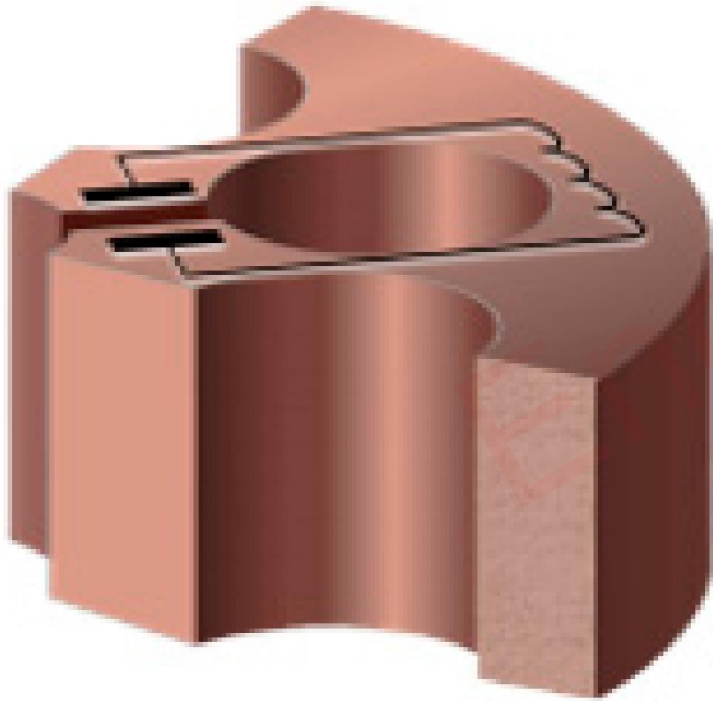
- ❑ **Electron Emission:** The heated cathode emits electrons, which are accelerated towards the anode by an applied electric field.
- ❑ **Magnetic Field Interaction:** The applied magnetic field causes the electrons to spiral and move in circular paths.
- ❑ **Resonant Cavities:** The spiraling electrons pass by the resonant cavities, inducing oscillations at microwave frequencies.
- ❑ **Energy Extraction:** The oscillating electromagnetic fields in the cavities couple to an output waveguide or antenna, producing a coherent microwave signal.

Filaments to
Supply Cathode

Equivalent Circuit of Magnetron Oscillator

- ❑ Magnetron Cavities are forming equivalent model of L {Inductor} and C {Capacitor} parallel combination [Tank Circuit].
- ❑ Resonance occurs at Frequency F.

$$F = \frac{1}{2\pi\sqrt{LC}}$$



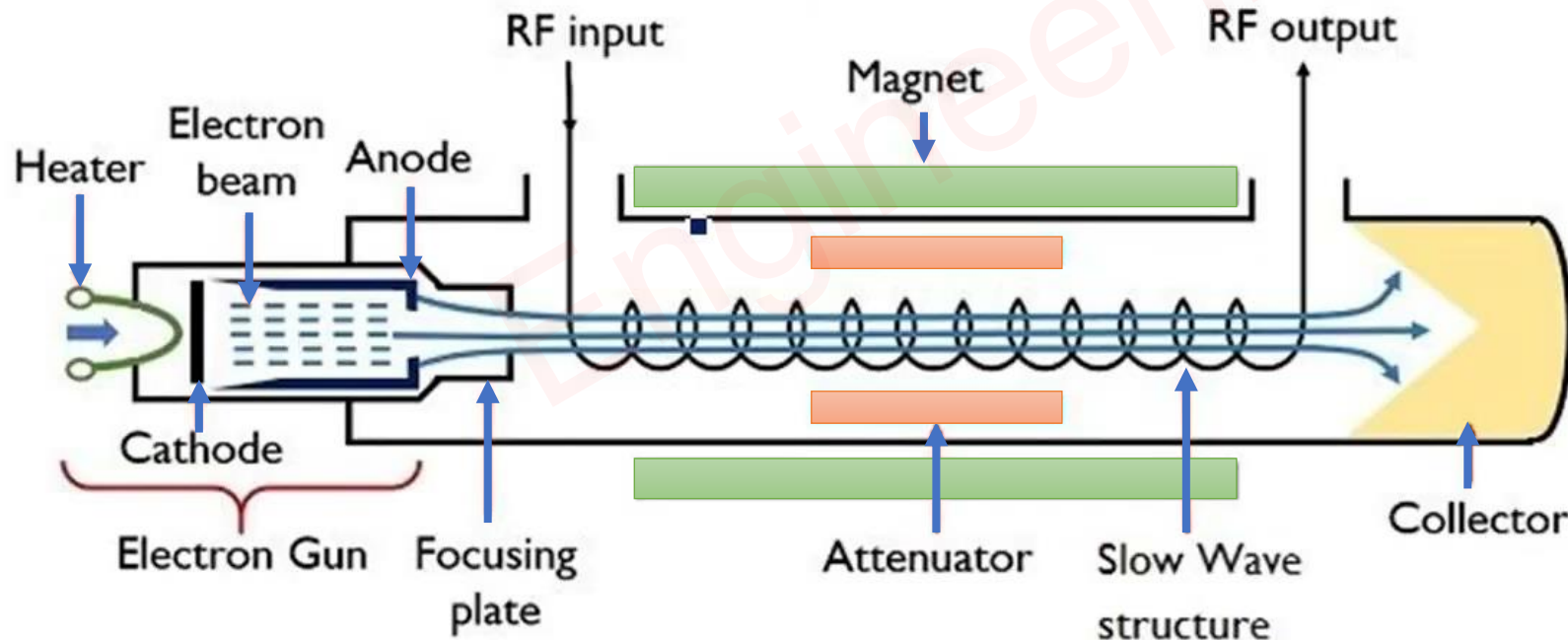
Applications of Magnetron Oscillator

- **Microwave Ovens:** Magnetrons generate microwaves to heat and cook food.
- **Radar Systems:** Used in military and civilian radar for detecting objects and measuring distances.
- **Communication Systems:** In some older communication systems for long-range signal transmission.
- **Industrial Heating:** Used in applications requiring non-contact heating, such as drying and melting materials.

Travelling Wave Tube

Basics of Travelling Wave Tube

- ❑ Travelling Wave Tube is a Microwave tube that amplifies Microwave/RF Signals.
- ❑ It belongs to the category of Linear beam tubes, such as Two Cavity Klystron.
- ❑ Travelling Wave Tube offers Wideband Frequency Amplification compared to other tubes.
- ❑ Travelling Wave Tube accounts for 50% of the total Microwave tubes.
- ❑ Travelling Wave Tube is commonly employed in radar, satellite communications, and other applications requiring high-power, high-frequency signal amplification.

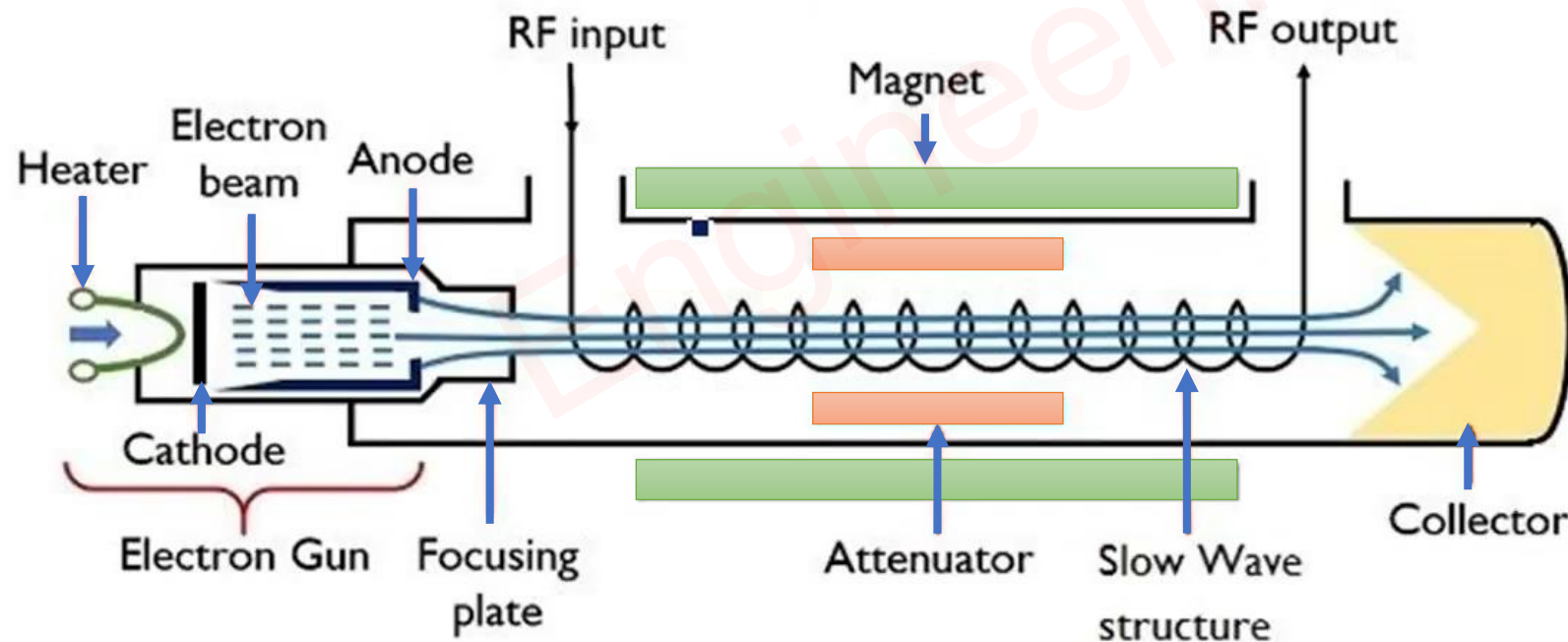


Difference Between TWT and Klystron

Parameters	Travelling Wave Tube	Klystron
Structure	❖ Slow Wave Structure {Helical Winding}	❖ Resonant Cavities
RF Interaction	❖ Continuous Interaction from RF Input to RF Output	❖ RF Input Interaction in Buncher Cavity Gap.
Bandwidth	❖ Wide Bandwidth	❖ Narrow Bandwidth
Gain	❖ 20 to 70 dB	❖ Around 60 dB
Noise Figure	❖ Lower	❖ Higher
Linearity	❖ Better	❖ Poor {It adds distortion in Amplified signal}
Efficiency	❖ Lower	❖ Higher
❖ Summary ☐ TWTs are preferred for applications needing wide bandwidth and moderate to high power. ☐ Klystrons are chosen for their high power output and efficiency in narrowband applications.		

Structure & Working of Travelling Wave Tube

- ❑ **Electron Gun & Collector:** An electron gun at one end of the tube emits a beam of electrons and the Collector at the other end receives an electron beam.
- ❑ **Slow-Wave Structure:** The electron beam travels through a structure called a slow-wave structure (often a helix or a coupled-cavity structure) that slows down the RF signal to match the velocity of the electrons.
- ❑ **RF Signal Interaction:** As the RF signal and electron beam travel together, the RF signal induces velocity modulation in the electron beam. This modulation results in the bunching of electrons. Due to the bunching of electrons, RF input is amplified at RF Output.



Parameters of Travelling Wave Tube

- ❑ Frequency Range: 300MHz to 50GHz
- ❑ Power Gain: 40 dB to 70 dB {Usually 60 dB}
- ❑ Power Range: Few Watts to Mega Watts
- ❑ Bandwidth: 10% to 20%

Applications of Travelling Wave Tube

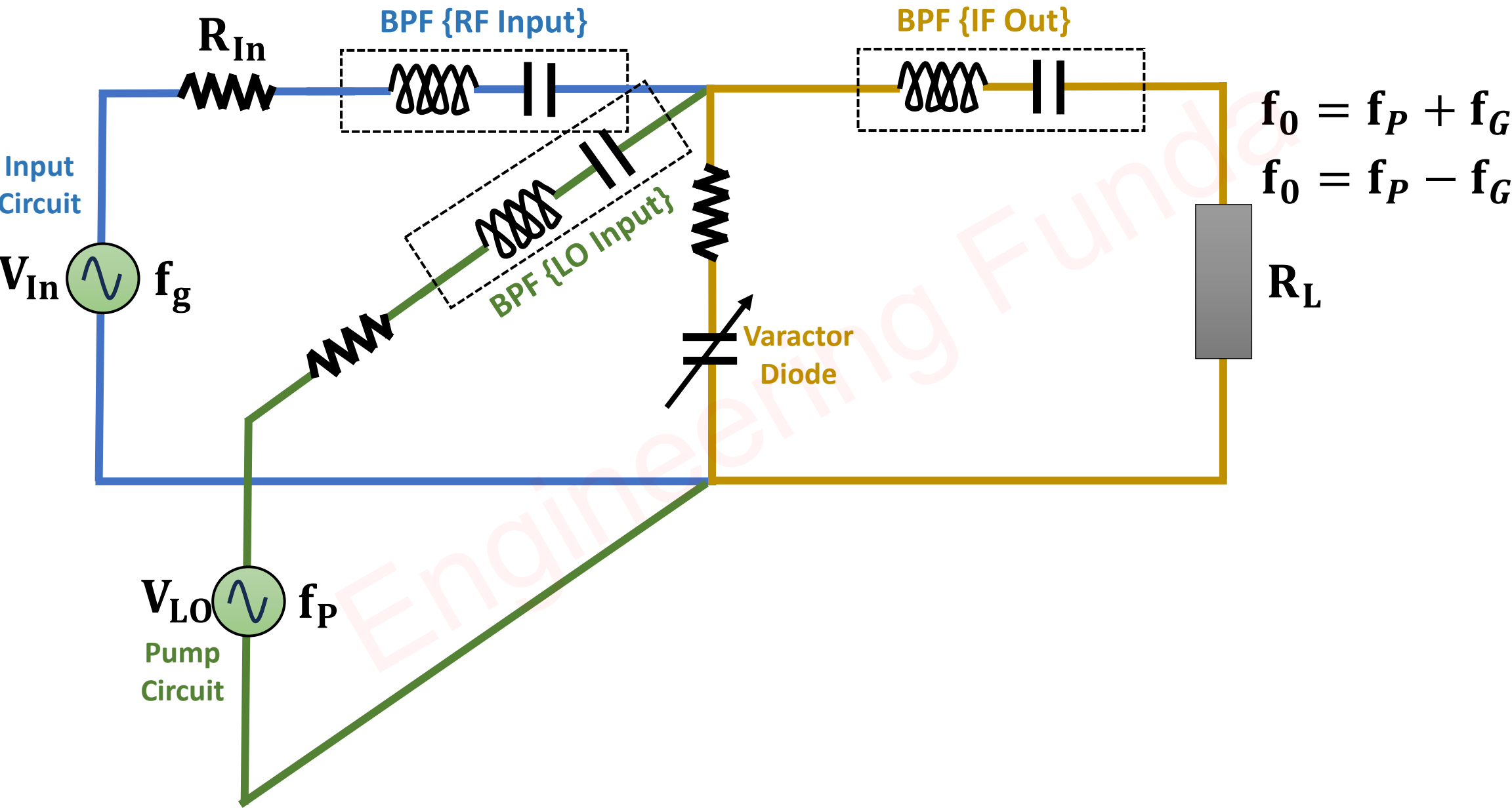
- ❑ **Radar Systems:** They are used in radar transmitters to amplify the radar signal, ensuring it can travel long distances.
- ❑ **Satellite Communication:** TWTs are widely used in satellite transponders to amplify signals for transmission to and from satellites.
- ❑ **Electronic Warfare:** TWTs are employed in electronic warfare systems for jamming and other signal amplification tasks.
- ❑ **Broadcasting:** High-power television and radio transmitters often use TWTs for signal amplification.

Parametric Amplifier

Basics of Parametric Amplifier

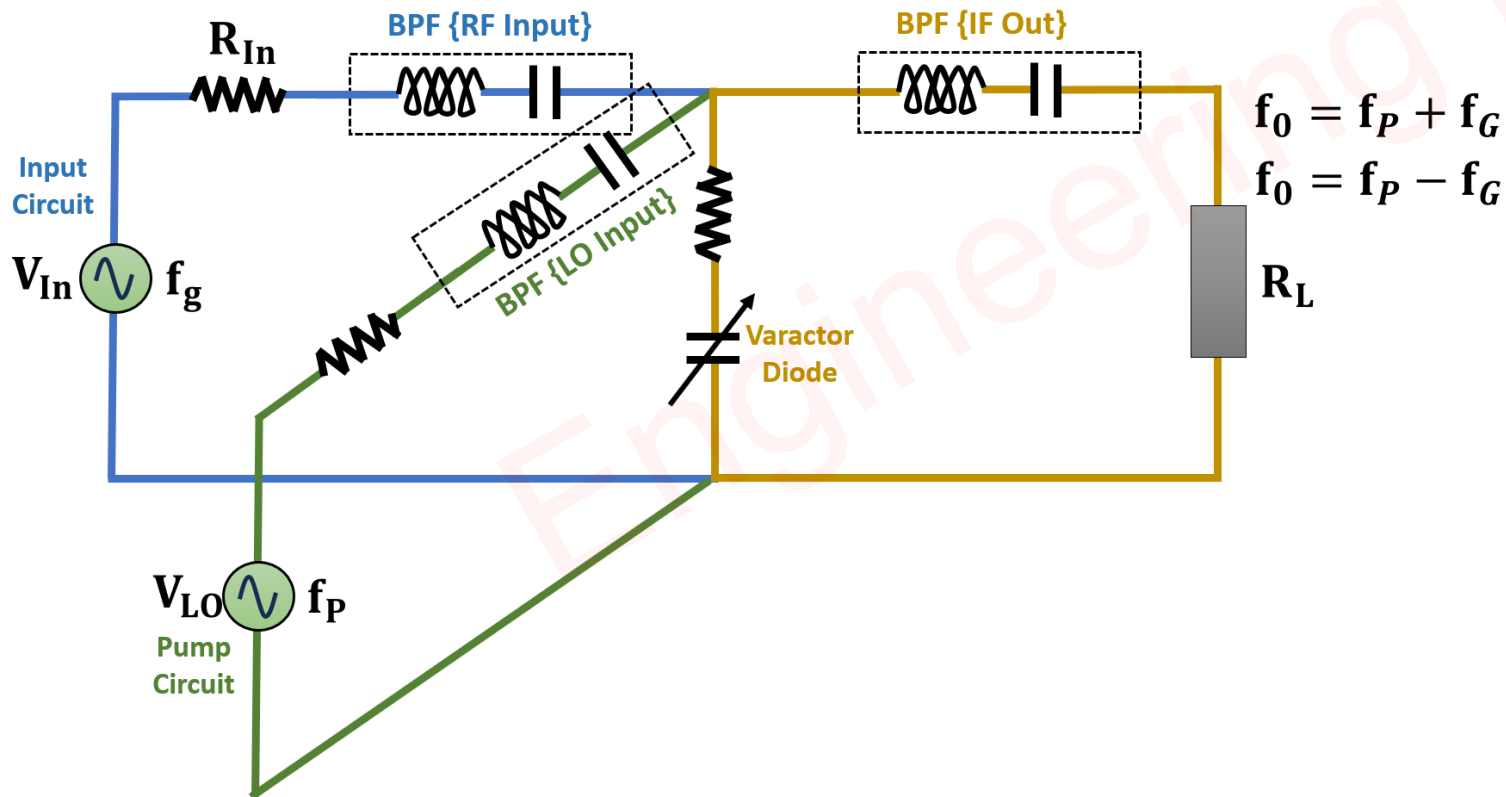
- ☐ Parametric Amplifier is a Highly Sensitive Low Noise Amplifier. {Noise Figure of 1-2dB}
- ☐ Parametric Amplifier is used for Ultrahigh Frequency and Microwave Frequency. {1GHz to 200GHz}
- ☐ Parametric Amplifier Utilizes active elements like Inductor and Capacitors.
- ☐ By Changing the Reactance, it amplifies signals using the pump Circuit.
- ☐ To Change the Reactance, the Varactor Diode is most commonly used as a variable reactor.

Circuit of Parametric Amplifier



Working of Parametric Amplifier

- At output, signal frequency is $f_p + f_G$ or $f_p - f_G$.
- As per Manley-Rowe, Maximum Gain $G = \frac{f_0}{f_G} = \frac{f_p + f_G}{f_G} = 1 + \frac{f_p}{f_G}$.
- This calculation is theoretical and practically Gain will be lower than it.
- If Output frequency is $f_p + f_G$, then parametric amplifier works as Up Converter and works as Amplifier.
- If Output frequency is $f_p - f_G$, then parametric amplifier works as Down Converter and works as Attenuator.



Advantages of Parametric Amplifier

- ❑ Noise Figure: Due to less resistance, the Range of Noise Figure is from 1 to 2dB.
- ❑ Frequency Range: It covers a wide range of frequencies {1GHz to 200GHz}.
- ❑ Gain: High Gain {From 20dB to 80dB}

Disadvantages of Parametric Amplifier

- ❑ Bandwidth: Low bandwidth due to a tuned circuit, but it can be increased by stagger tuning.
- ❑ Complex Design: Precise Implementation is Complex.

Applications of Parametric Amplifier

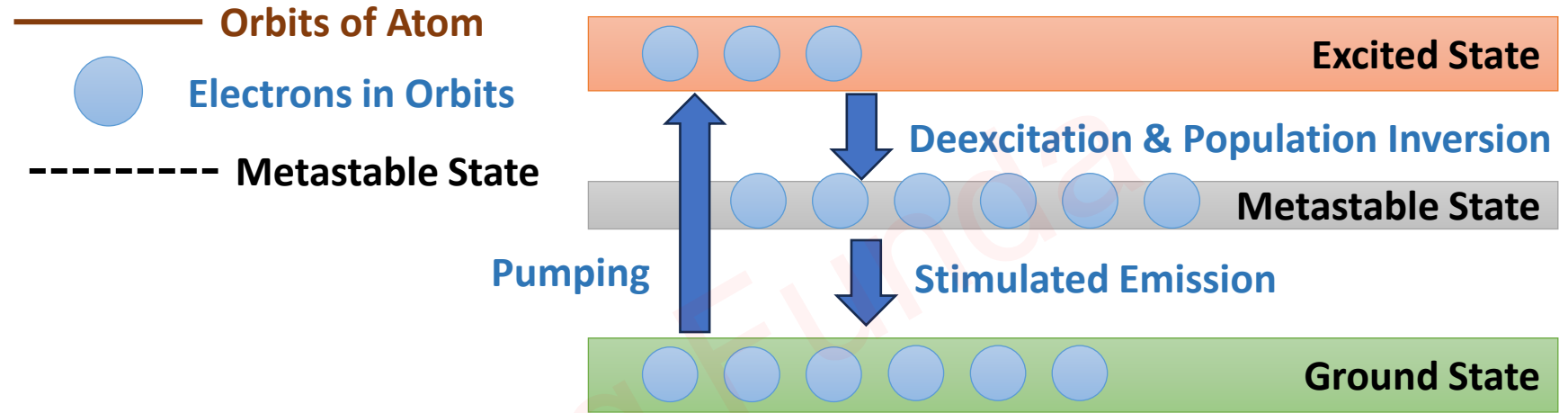
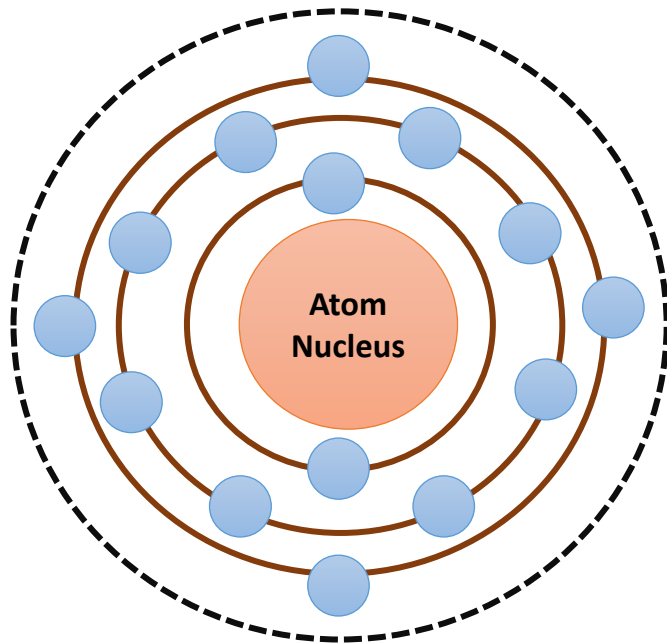
- ❑ LNA – Low Noise Amplifier
- ❑ Optical Communication
- ❑ Radio Astronomy

MASER

Basics of MASER

- ❑ MASER is **M**icrowave {**M**olecular} **A**mplification **S**timulated **E**mission **R**adiation.
- ❑ MASER is used to generate Microwave Frequency and it is also used to generate Infrared Frequency.
- ❑ MASER working is based on the principle of LASER, MASER works based on **P**opulation **I**nversion and **S**timulated **E**mission.

Working of MASER



- By Pumping Energy, Electrons from the Ground State go towards the Excited State.
- After that electrons from the Excited state move toward the Metastable state {At the Metastable state many electrons are gathered}, It is referred to as population inversion.
- By Stimulated Emission, Electrons from the Metastable state moved to the Ground state, which generates a Microwave signal.
- Spontaneous Emission happens with Excited Electrons with Life span of 10^{-8} Sec. Stimulated Emission happens with Metastable state with life space of 10^{-3} Sec.

Materials of MASER

- ☐ Ruby
- ☐ Rubidium
- ☐ Ammonia
- ☐ Chromium
- ☐ Gadolinium Ions

Applications of MASER

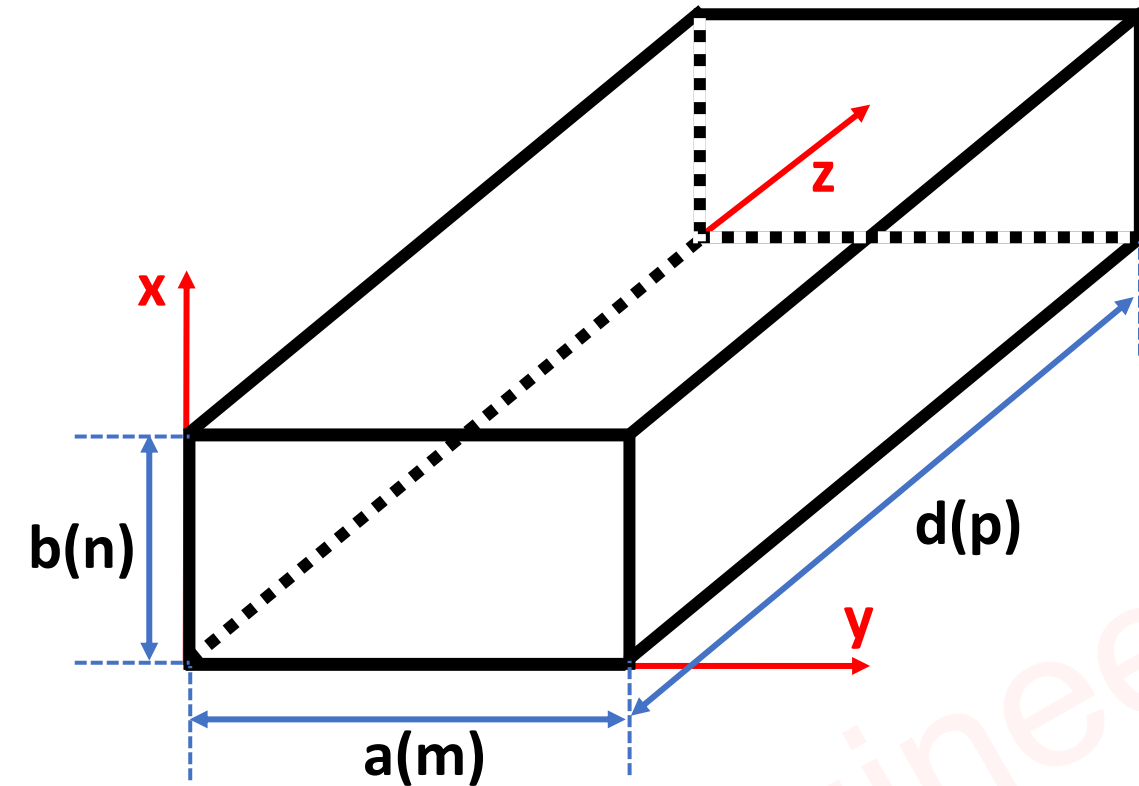
- ☐ Communication {Long Range Communications}
- ☐ Radio Astronomy
- ☐ Molecular Spectroscopy

Cavity Resonator

Basics of Cavity Resonator

- ❑ A Cavity Resonator is a Hollow Metallic enclosed tube that exhibits resonance behavior when excited by an EM Field.
- ❑ A Cavity Resonator is a structure that confines electromagnetic waves, such that standing wave patterns are formed within the cavity.
- ❑ Cavity Resonators are mainly used in Microwave Oscillators, Tuned Amplifiers, Frequency Meters & Phased Equalizers.
- ❑ Microwave Signals have Lumped Elements {Lumped Elements restricts Quality Factor Q }.
- ❑ Microwave Signals have small Physical Dimensions {Small Dimensions Limits V & I }.

Resonance Frequency of Cavity Resonator



□ Resonance Frequency of Cavity Resonator is given by:

$$f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

■ Where: m , n , and p defines the mode of resonance

• Rectangular Cavity Resonator:

- Dimensions: $a = 10$ cm, $b = 5$ cm, $d = 2$ cm
- Fundamental Mode ($m = 1, n = 0, p = 0$):

$$f_{100} = \frac{c}{2a} \approx \frac{3 \times 10^8 \text{ m/s}}{2 \times 0.1 \text{ m}} = 1.5 \text{ GHz}$$

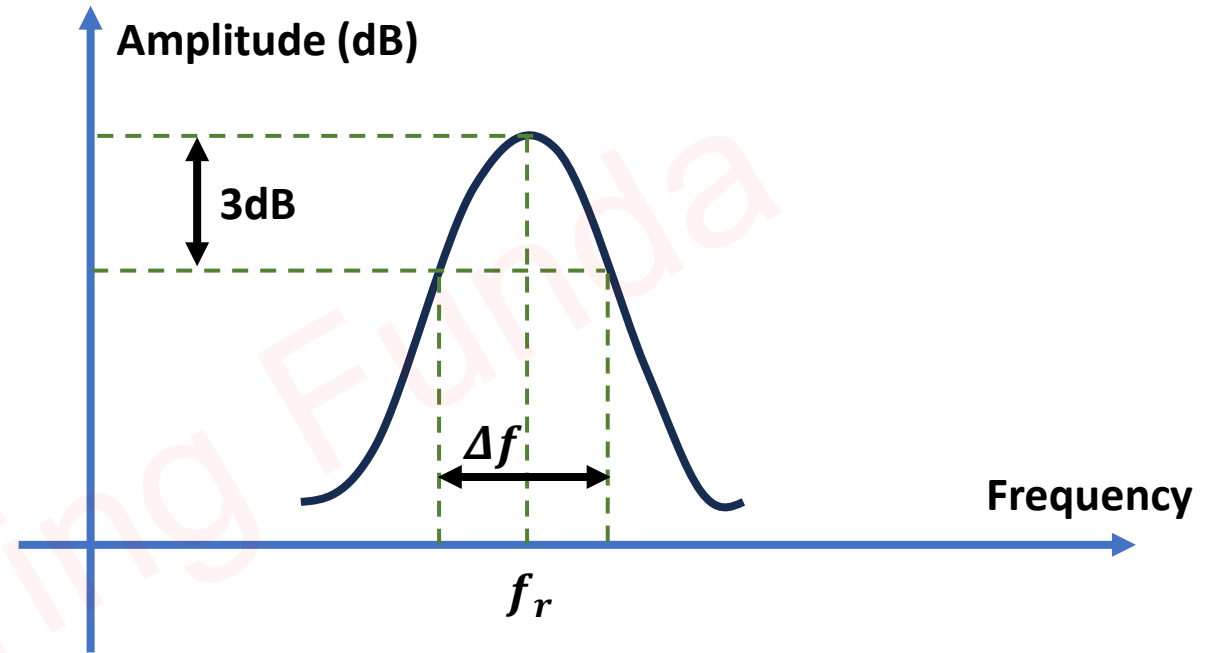
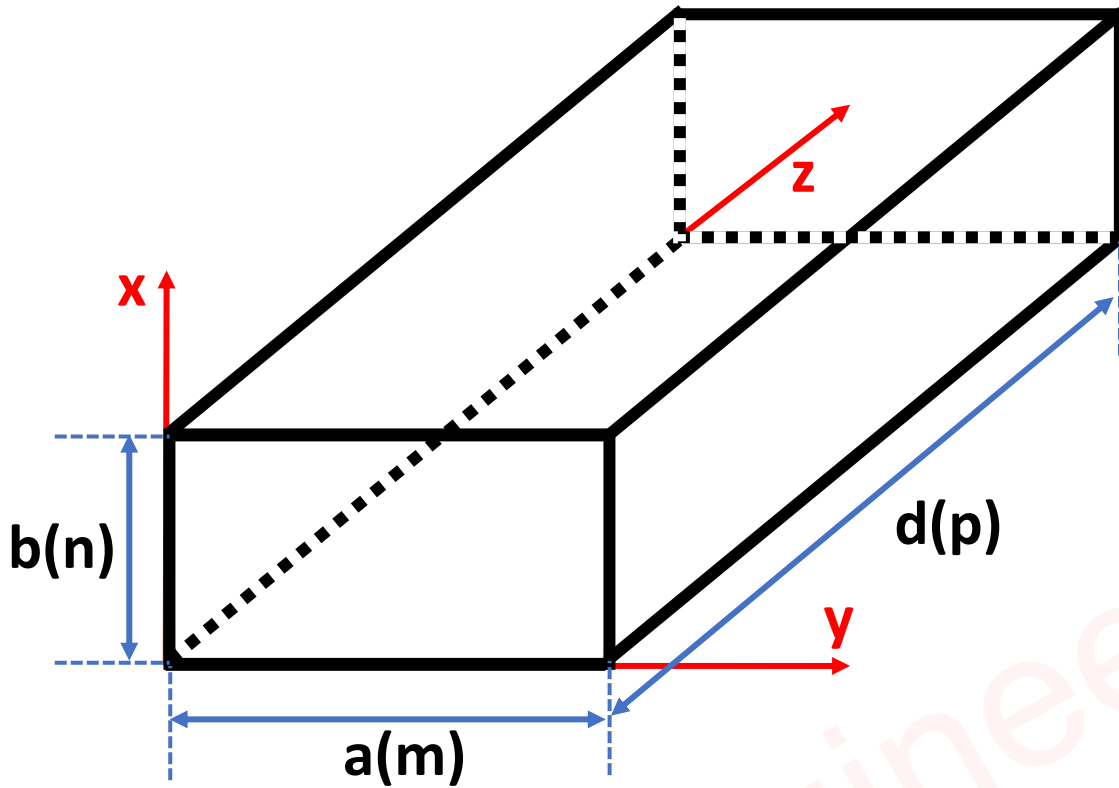
□ Only TE and TM modes are possible in waveguide Cavity Resonator.

□ TEM mode is not possible in waveguide Cavity Resonator.

□ Resonance happens at standing pattern, where dimensions $\{a, b, \text{ and } d\}$ are multiple of $\lambda_g/2$.

□ So, Dominant Mode happens at dimension $d = \lambda_g/2$.

Quality Factor of Cavity Resonator



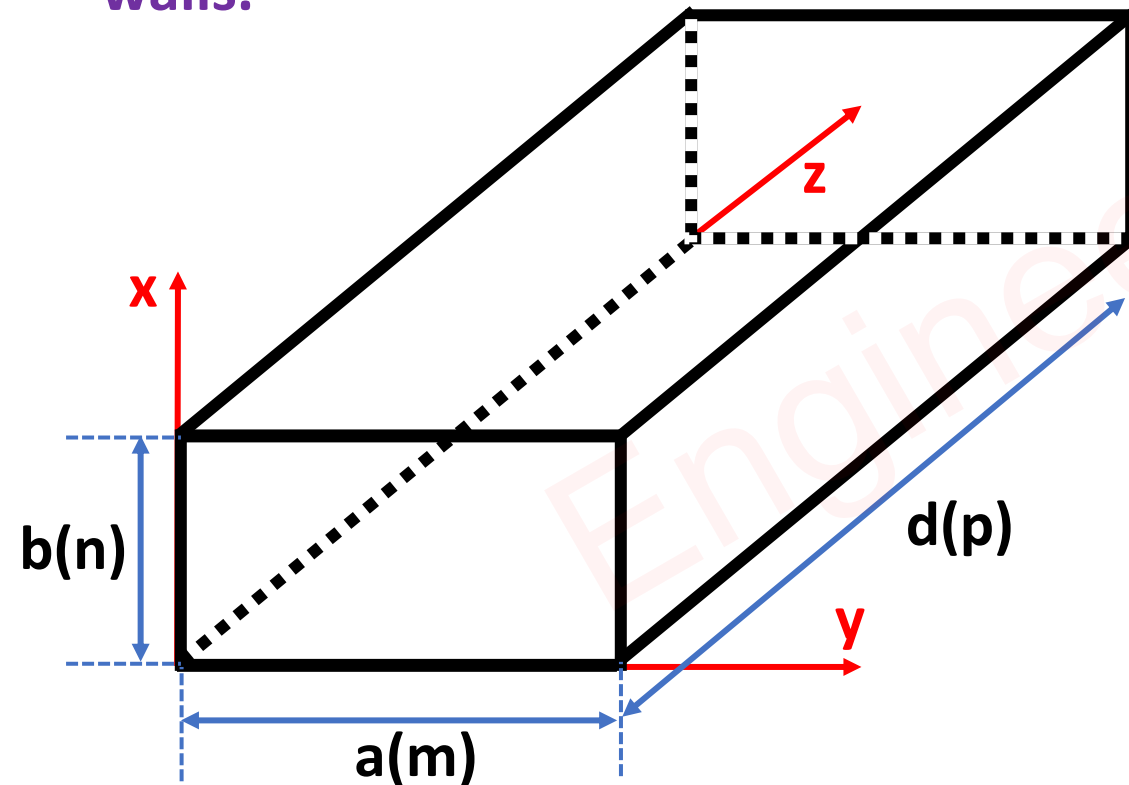
□ Quality Factor of Cavity Resonator is given by:

$$Q = \frac{f_r}{\Delta f}$$

- Where: f_r is the resonance frequency
- Δf is 3dB bandwidth

Mode Structure and Boundary Condition of Cavity Resonator

- ❑ **Mode Structure:** The specific patterns of the standing waves within the cavity. These patterns are defined by mode numbers $\{TE_{mnp}$ or $TM_{mnp}\}$, which correspond to the number of half-wavelengths fitting into the cavity dimensions.
- ❑ **Boundary Conditions:** The walls of the cavity impose boundary conditions that determine the possible modes. For instance, in a metallic cavity, the electric field must be zero at the walls.



❖ For wave Propagation, Maxwell's Equation is

$$\Rightarrow \Delta^2 E + k^2 E = 0$$

- Where: k is propagation constant

Applications of Cavity Resonator

- ❑ **Oscillators:** Provide stable frequencies for signals.
- ❑ **Filters:** Used to select specific frequencies from a broad spectrum.
- ❑ **Microwave Communication:** Essential components in microwave transmitters and receivers.
- ❑ **Spectroscopy:** Used in devices like masers and lasers for precise frequency control.

Why Conventional Tubes are Not used at Microwave Frequencies?

❑ Introduction of Lumped Elements

- At Microwave Frequencies, Many Lumped Elements {Inductance, Capacitance, Transconductance and Resistance} are added.

❑ Interelectrode Capacitance and Lead Inductance Effect

- The Input Impedance of the Tube is given by the following expression:

$$Z_{in} = \frac{1}{\omega^2 LCg} - j \frac{1}{\omega^3 LCg^2}$$

- Where: ω is the operating frequency
- L is Lead Inductance
- C is Interelectrode Capacitance
- g is Transconductance
- At Microwave Frequencies, Input Impedance decreases which leads to a decrease in the power handling capacity of the tube.

□ Transit Angle Effect

- The transit angle for tube is given by the following expression:

$$\theta_g = \frac{\omega d}{V_0}$$

- Where: ω is the operating frequency
- d is a separation between the cathode and the grid
- V_0 is applied DC Voltage
- At Microwave Frequencies, the Transit angle becomes significantly large which leads to phase distortion in signals.

□ Gain Bandwidth Product Limitation

- The Gain Bandwidth Product for tube is given by the following expression:

$$\text{Gain} \times \text{BW} = \frac{g_m}{C}$$

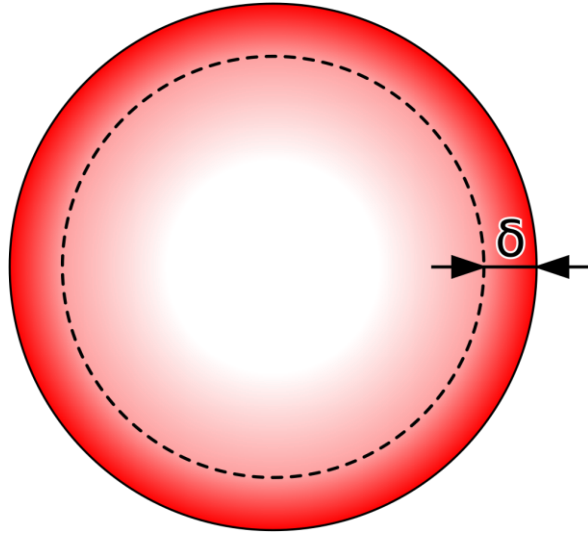
- Where: g_m is Tube Transconductance
- C is Tuning Element
- At Microwave Frequencies, for higher gain Bandwidth is limited.

□ Radiation Losses Effect

- At Microwave Frequencies, Simple Lead or wire acts as Antenna.

□ Skin Effect

- At Microwave Frequencies, Current will flow on the skin of conductors.
- Skin depth δ can be calculated by:



$$\delta = \sqrt{\frac{2\rho}{\omega\mu}}$$

- Where: ω is the operating frequency
- ρ is a resistivity of Material
- μ is magnetic permeability of material

□ Dielectric Losses

- At Microwave Frequencies, there will be loss of signal in dielectric materials.